

Bayesian regression for language scientists

Lecture 1

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LOT2026 Summer School

This course

- Today: Bayesian regression in language sciences
- Tuesday: Bayesian linear mixed models
- Wednesday: Bayesian generalized linear mixed models
- Thursday: Bayesian phylogenetic models
- Friday:
 - Problems of using Bayes factors
 - Best practices
 - Discussion of your projects (prepare a 1-2 minute description of how you could use Bayesian regression in your own research)

Course materials

<https://github.com/levshina/LOT2026>

Software

- R and RStudio
- R package brms
 - user-friendly syntax, similar to lme4
 - uses Stan for compiling models in C++
- You need to configure your computer so that it can use C++
 - See `software_instructions`
 - Make sure you can reproduce the example

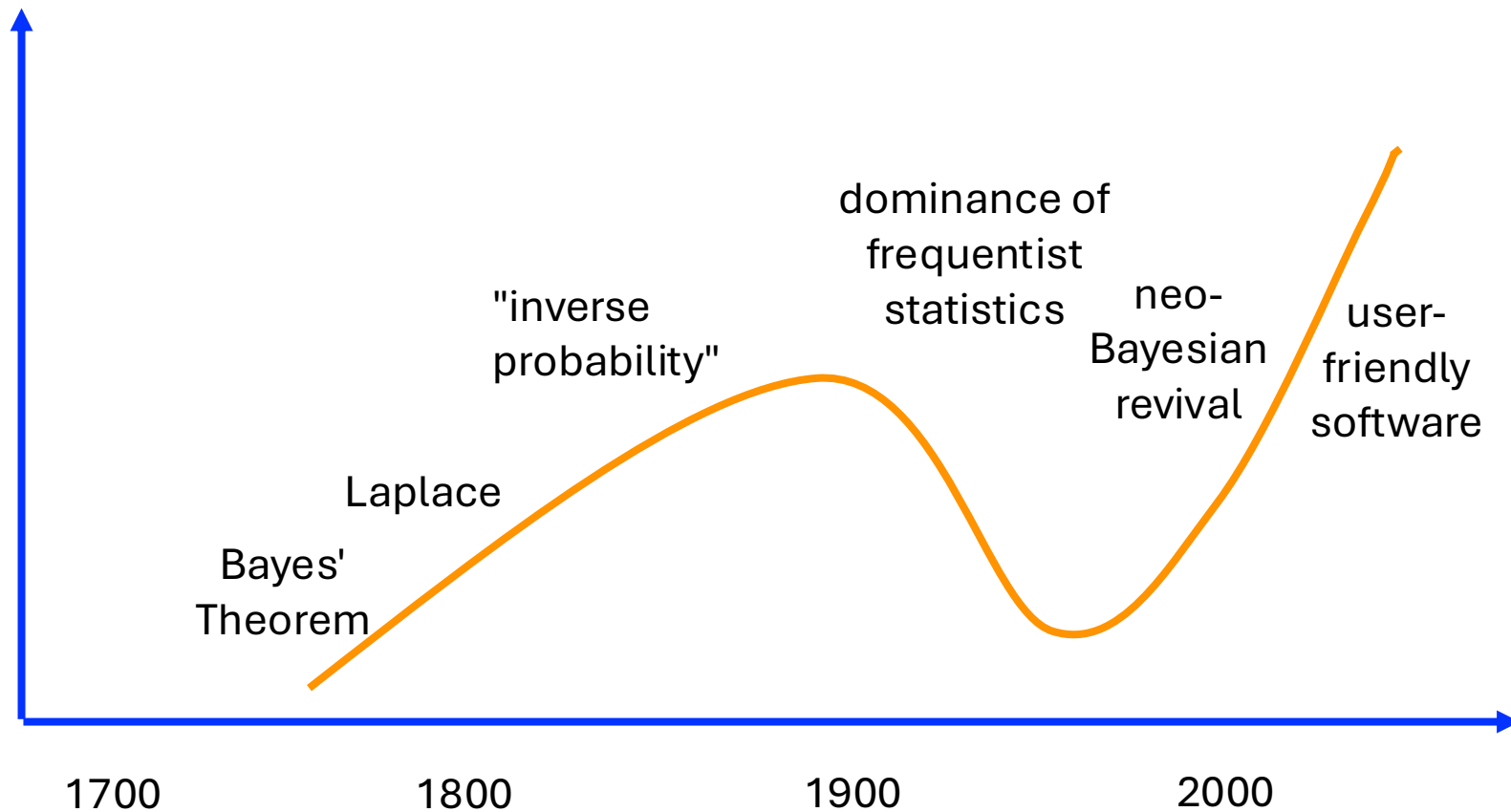
Outline of today's lecture

- A short history of Bayesian statistics
- What makes Bayesian statistics special?
- How is Bayesian regression used in language sciences?
- Should I learn Bayesian statistics?

Adjective "Bayesian"

- First used by R. Fisher in a derisive comment
- Adopted by Bayesians in the 1950s
- Now used in a neutral sense

Bayesian statistics timeline



Reasons for decline

- Bayesian statistics is associated with the idea that probabilities are subjective beliefs – doesn't fit the positivist, objectivist spirit.
- Bayesian statistics was not feasible for complex questions due to computational limitations, which were resolved only later (MCMC algorithms).
- Personal influence of Fisher (applications in agriculture in U.S. land grant institutions) and Neyman (Berkley).

Neo-Bayesian revival

- In 1975, British statistician Dennis Lindley predicted that the 21st century would be Bayesian.
- The proportion of papers related to Bayesian statistics in five major statistical journals rose from under 5% in 1990 to approximately 20% by 2015 (Lenhard 2022).
- 1,500 Bayesian articles in Psychology published between 1990 and 2015, with a clear upward trend (van de Schoot, Winter, Ryan, Zondervan-Zwijnenburg & Depaoli 2017).
- Bayesian methods are still not dominant, but they have gained a significant traction.

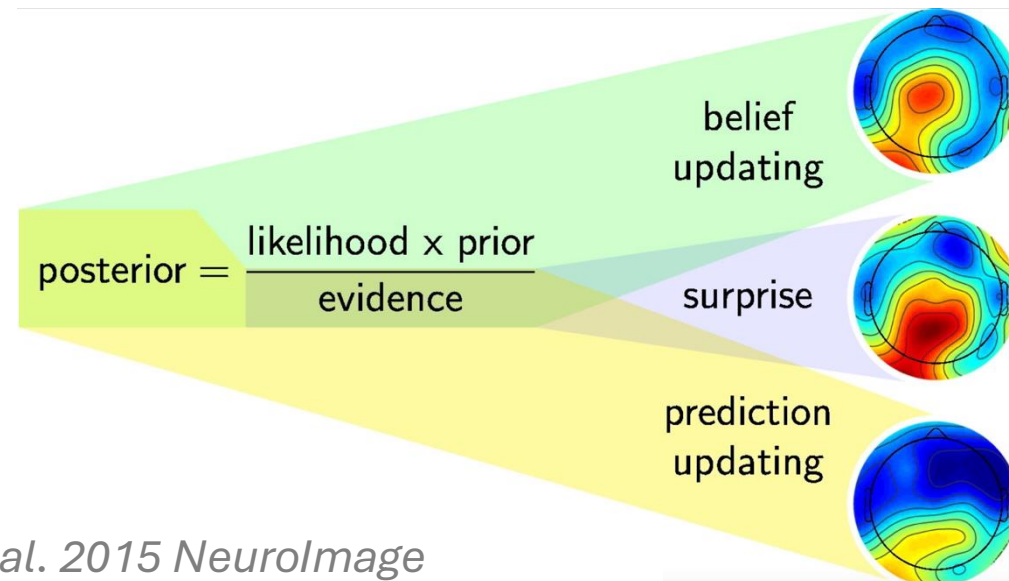
Causes: Development of tools

- Development of powerful computers and efficient algorithms based on the Monte Carlo method (John von Neumann, **Stan**islaw Ulam, Nicholas Metropolis)
- 1989: Bayesian inference Using Gibbs sampling (BUGS)
- 1997: WinBUGS
- 2007: platform-independent JAGS in C++, later rjags in R
- 2013: Stan in C++ (2013) and Rstan: Hamiltonian Monte Carlo and its variant No-U-Turn sampler (NUTS)
- 2017: brms – a convenient wrapper for Stan in R; makes Bayesian regression more accessible to people who already know lme4



Causes: Bayesian models of cognition

- Bayesian theories of perception, learning, reasoning and language processing in cognitive science (*Kruschke 2010*)
- Bayes rule provides a formalization of how humans use prior beliefs or expectations when processing new information and how these beliefs are updated (*Chater et al. 2010*).



Causes: Replicability crisis

- Highlighted deeply rooted issues with traditional statistics, especially with Null Hypothesis Significance Testing and p-values.
 - misinterpretation of results
 - questionable practices (p-value hacking, HARKing)
 - publication bias
- Countermeasures: promotion of open data and code, preregistration...
- Bayesian statistics emerged as a viable alternative to the traditional approaches.

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What are probabilities?

Frequentist statistics

Probabilities as long-run frequencies

If a coin is fair, and we toss the coin very many times, we expect to get the heads 50% of the times and the tails 50% of the times. The more times we repeat the experiment, the closer the observed proportions of heads and tails will be to 50%.

Bayesian statistics

Probabilities as beliefs

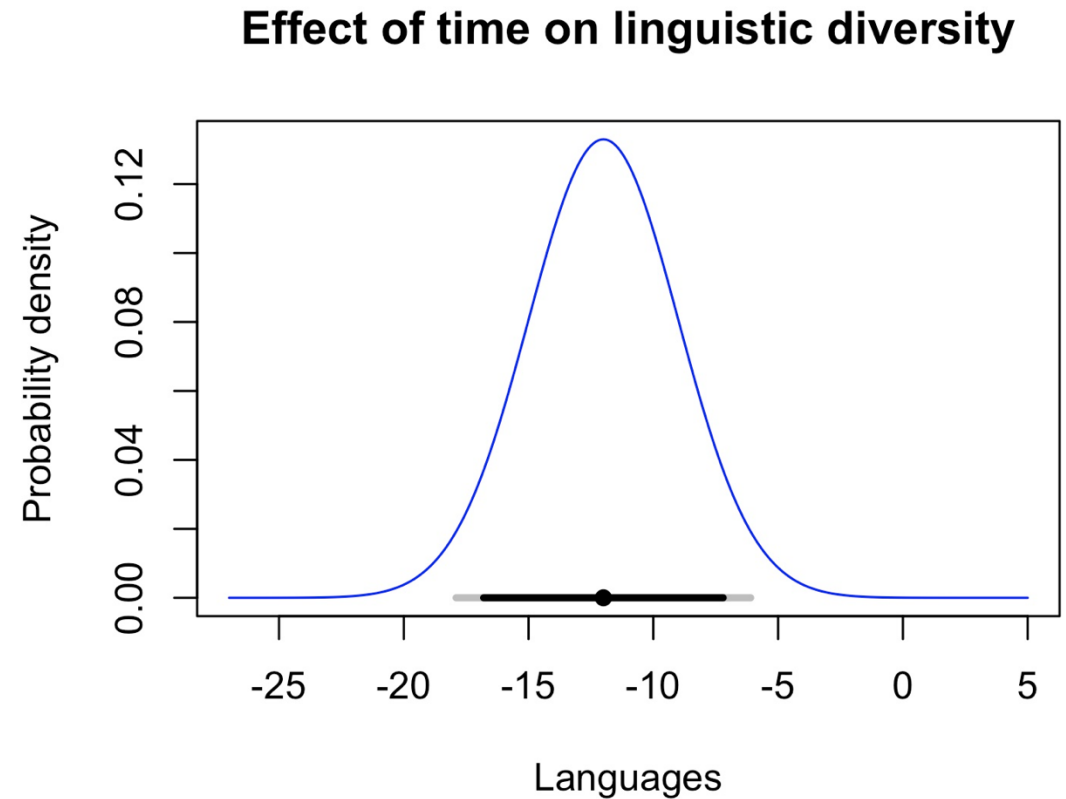
Beliefs - an umbrella term for expectations, opinions, assumptions, convictions... For every value of a parameter, one can think of its probability.

What are betas in regression?

Frequentist statistics	Bayesian statistics
Parameters as unknown but fixed constants A coefficient estimate in regression is one number.	Parameters as random variables represented as probability distributions A coefficient estimate is a probability distribution.

Illustration: loss of linguistic diversity

- Bromham et al. (2022): at least one language is lost per month if we do not intervene.
- An imaginary regression coefficient of Time (in years):



Basic terms

$$P(\theta | \text{data}) = P(\text{data} | \theta) * P(\theta) / P(\text{data})$$

- $P(\theta | \text{data})$ - the posterior probability of θ given the knowledge of the data. In other words, a posterior distribution is the degree of belief in every value of the parameter after considering the data.
- $P(\text{data} | \theta)$ - the probability of the data given θ , also known as the likelihood of θ for given data (*Fisher 1922*).
- $P(\theta)$ - the prior distribution, or the distribution of prior probabilities of θ without the knowledge of the data; in other words, the degree of belief in every specific value of the parameter before considering the data.
- $P(\text{data})$ - evidence, or marginal likelihood. It is usually a normalizing constant that is necessary to make sure that the posterior distribution integrates to one.

Posterior distribution $\propto$ Likelihood $\times$ Prior distribution

Example

- Effect of word surprisal in bits on regression-path duration

Researchers

Assertia



Brelax



Causy

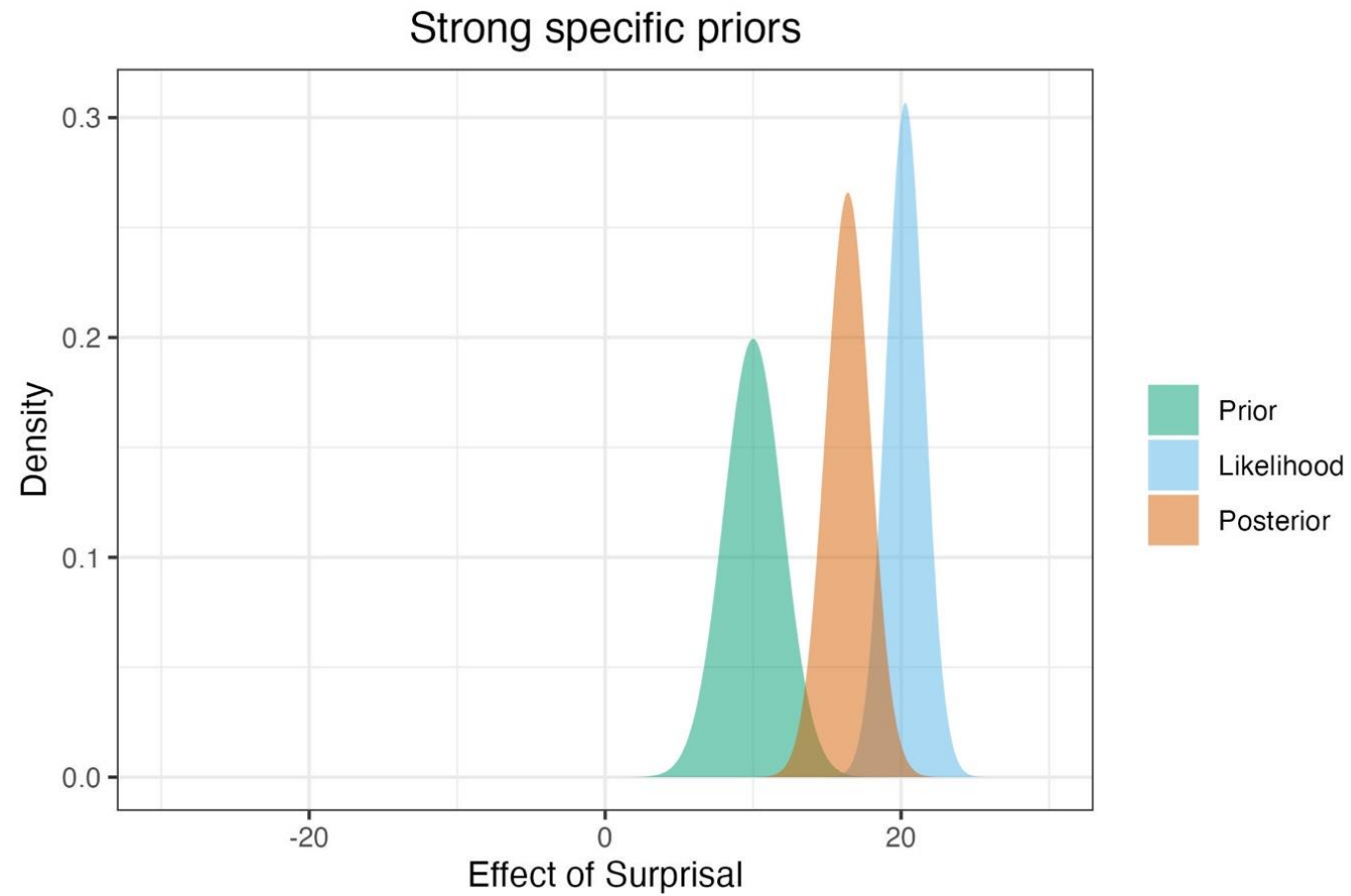


Informative priors

- Assertia has strong beliefs about the effect of surprisal: one bit of surprisal adds 10 ms of duration
- She uses informative, strong specific priors.



Assertia's results

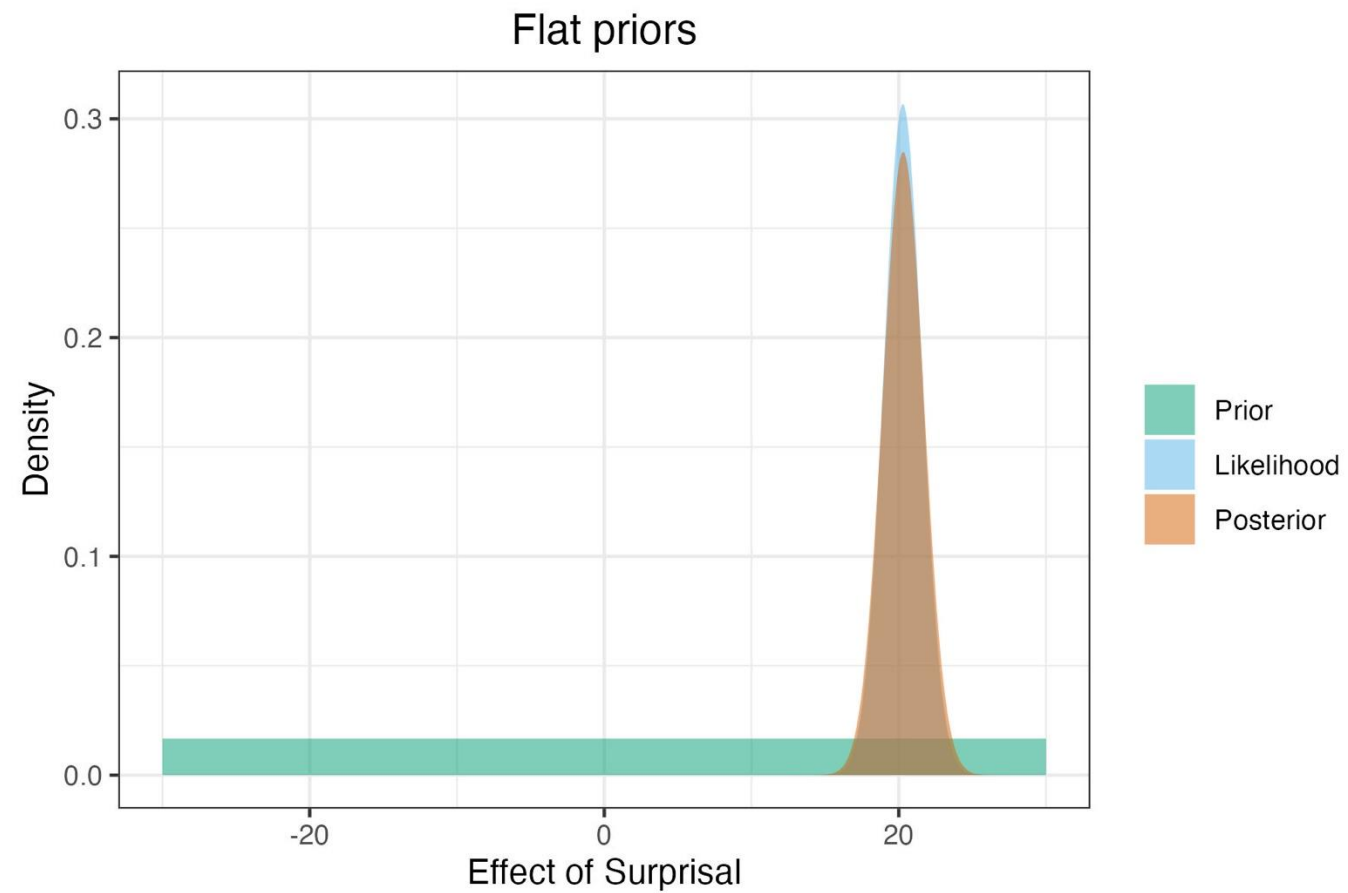


Brelax

- Brelax: all values are equally likely.
- He uses uniform (vague, diffuse, flat) priors



Brelax' results

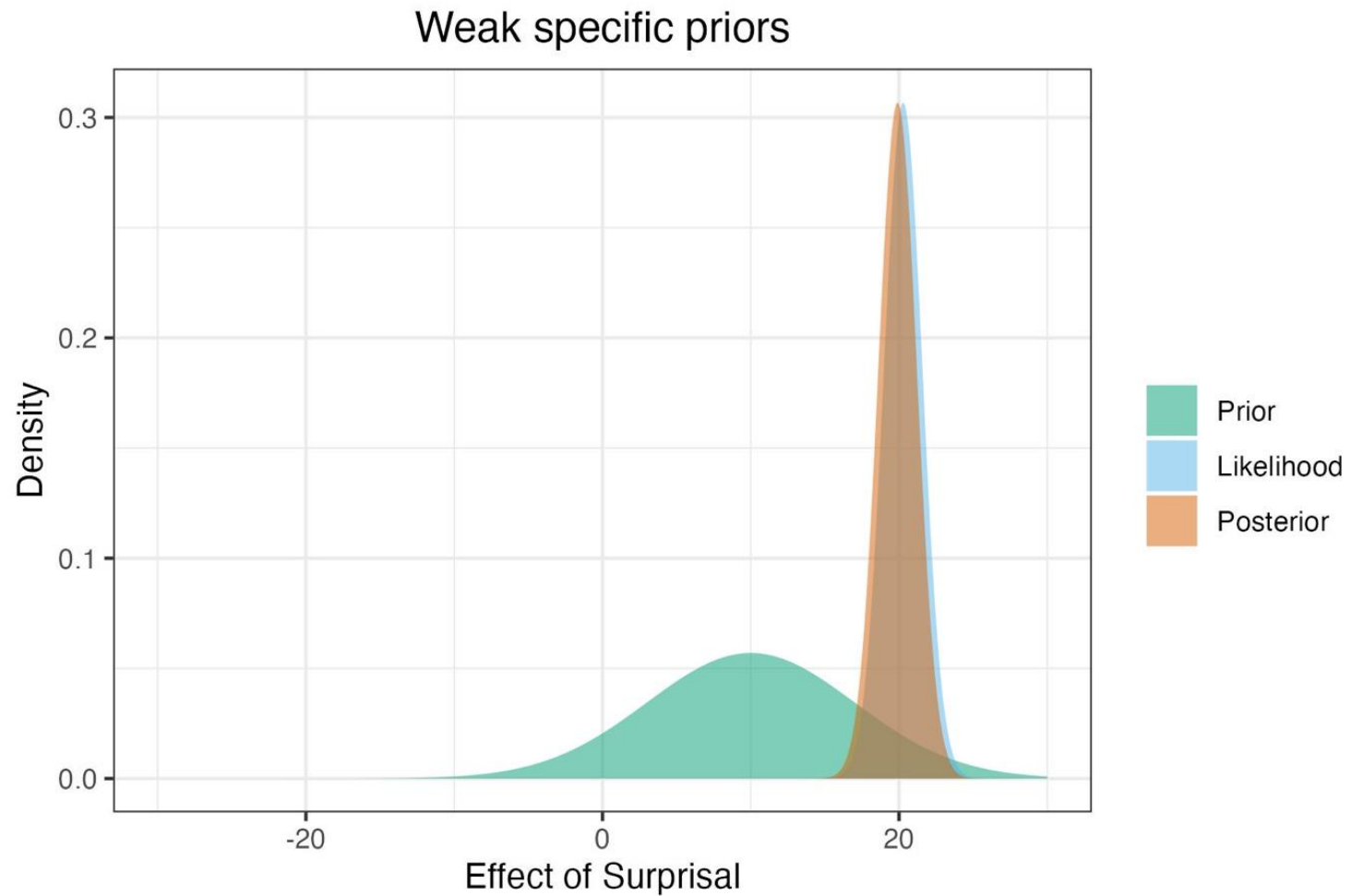


Causy

- Causy has specific priors around 10 ms, like Assertia, but these priors allow for more variation because the distribution is more spread out.
- Such priors are called weakly informative priors.

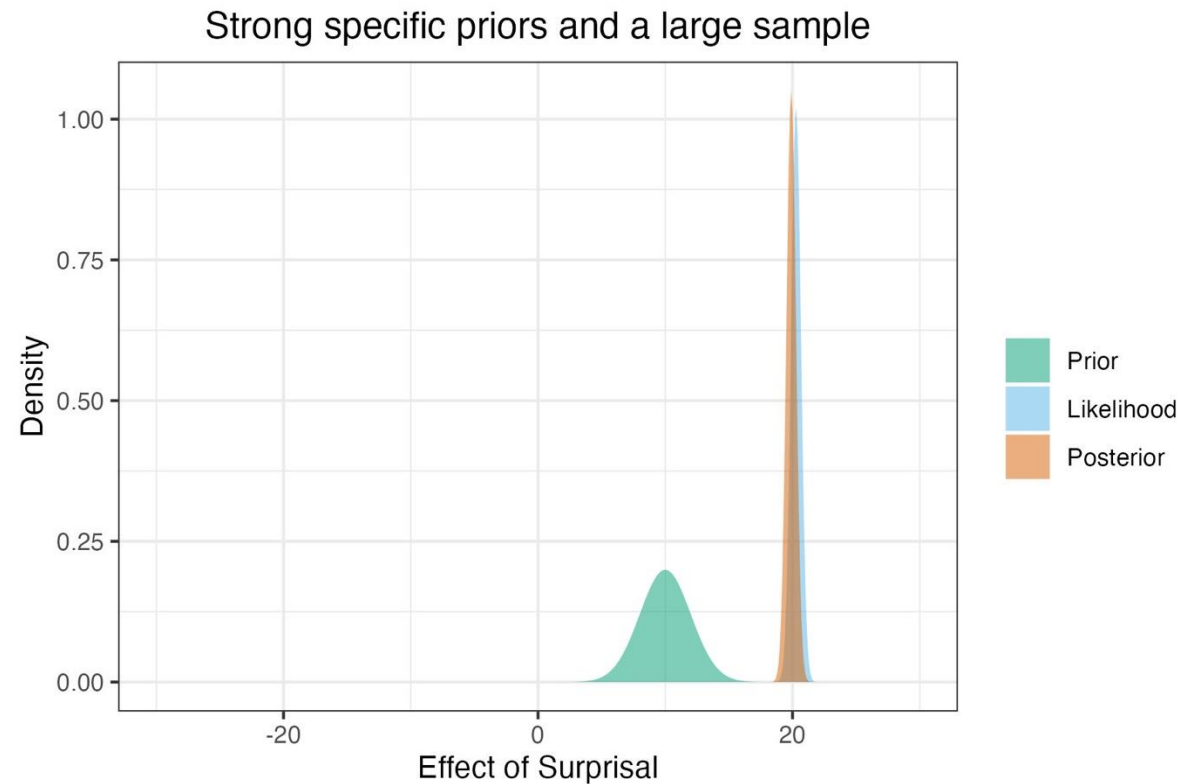


Causy's results



More data

- If Assertia gets some extra data and uses the same priors, this is what she obtains:



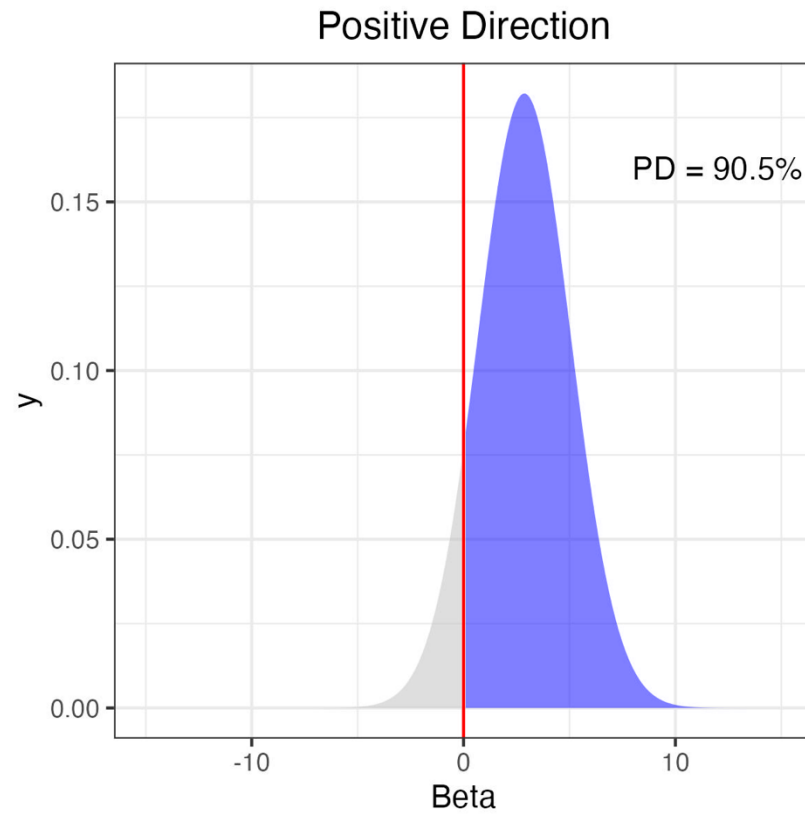
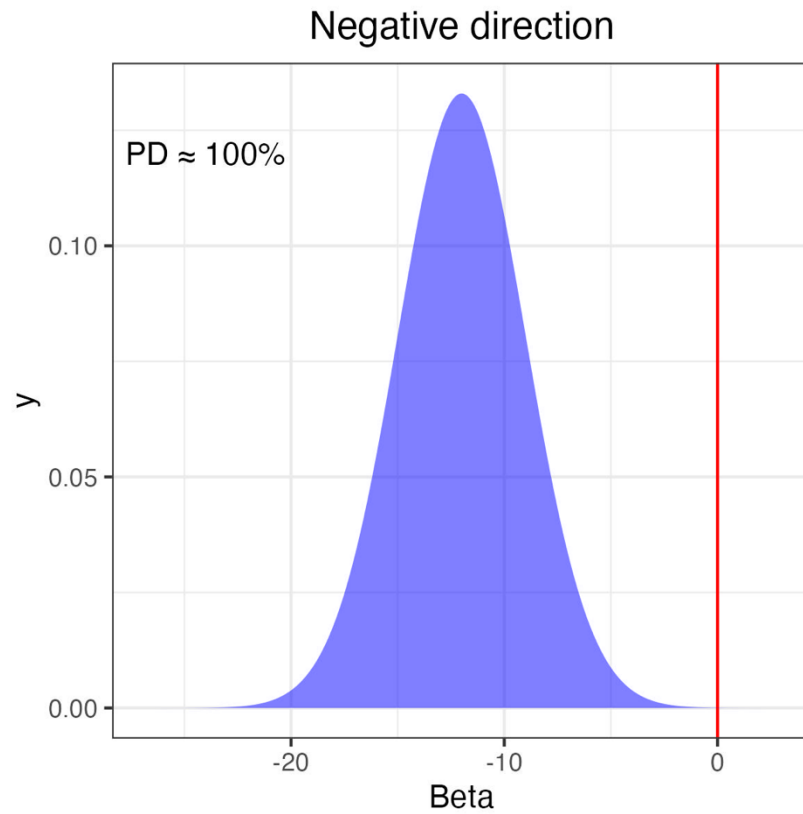
MCMC: a key ingredient

- Monte Carlo Markov Chains estimation enables us to estimate posterior probability distributions, which would not be possible otherwise in most practical cases.
- Markov Chains are models in which the probability of an event depends only on the state attained as the result of the previous event.
- Monte Carlo is a family of methods related to random sampling from a distribution.
- We can sample directly from the posterior without assuming any shape (beta, Gaussian, etc.)
- This can be done with Stan, a platform for high-performance statistical computation: <https://mc-stan.org/> , named after Stanislaw Ulam, one of the inventors of MCMC.

Hypothesis testing

Frequentist statistics	Bayesian statistics
Null hypothesis significance testing: Can we reject the null hypothesis? Tools: p-values, confidence intervals.	Testing the research hypothesis directly: Can we support the research hypothesis? Tools: ROPE, Probability of Direction Which hypothesis is more supported by the data? Tools: Bayes Factor, ROPE...

Probability of Direction



Intervals

Frequentist statistics

Confidence intervals:

If we calculate a 95% confidence interval in this way many times on repeated samples, the true mean will be between the boundaries 95% of the time.

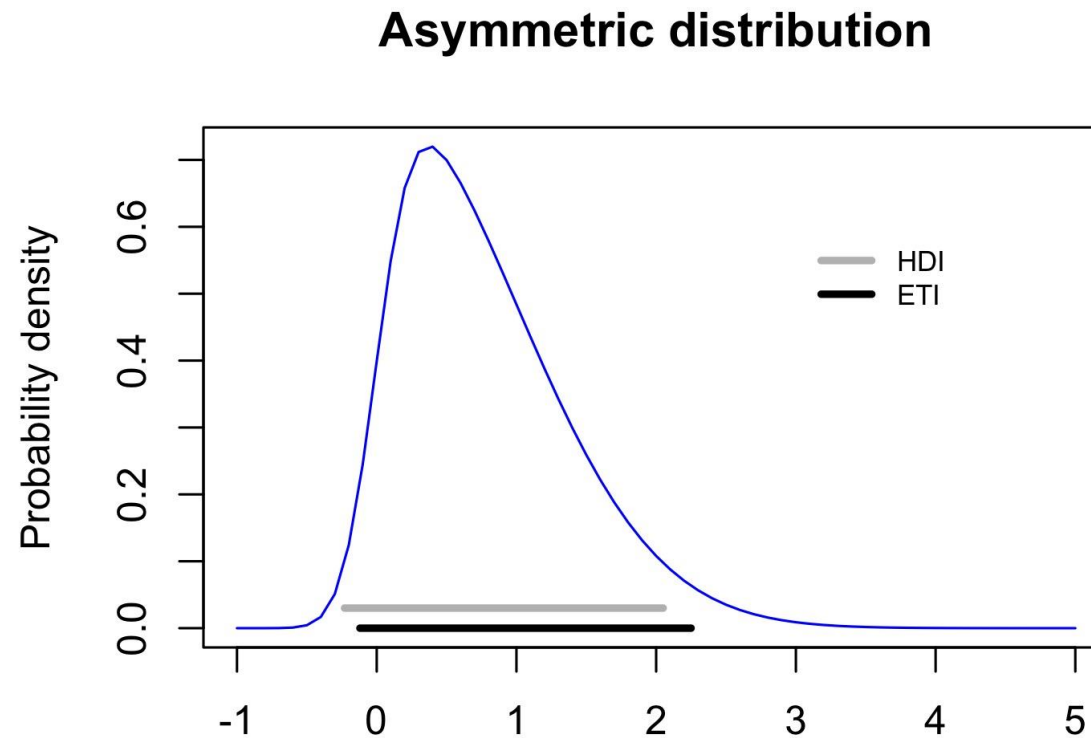
Bayesian statistics

Credible intervals:

the parameter of interest (e.g. a regression coefficient) falls in this interval with the probability of ... %

Different types: a) Highest Posterior Density
b) Equal-Tailed Interval

HDI vs. ETI



Misconceptions

- Bayesian statistics is completely different from frequentist statistics.
 - No, especially if flat or weakly informative priors are used. The results are normally very similar.
- The use of priors is subjective and therefore not scientific.
 - No, priors are usually based on common sense or on previous research. Sensitivity to priors should be analyzed and reported.
- Bayesian models don't need diagnostics.
 - Actually, there is more diagnostics to perform, related to MCMC chain convergence: R-hats, effective sample sizes, autocorrelations...
- Bayesian statistics is immune to misuse, unlike frequentist statistics.
 - Some tools (e.g., Bayes Factors) are prone to misinterpretation and misuse, e.g., BF-hacking:
 - "... surely, God loves the .06 nearly as much as the .05" (*Rosnow & Rosenthal 1989: 1277*)
 - "God would love a Bayes Factor of 3.01 nearly as much as a BF of 2.99." (*Konijn et al. 2015*)

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Bayesian regression in linguistics

- Vasishth et al. (2013) used JAGS for a meta-analysis of studies about the processing of Chinese relative clauses, in the form of a Bayesian multilevel regression.
- Bayesian multilevel linear models
 - Husain et al. (2014) on locality and expectation effect in Hindi data
 - Hofmeister and Vasishth (2014) on sentence processing based on English data
- Bayesian generalized linear models
 - Skrivner (2015), who used JAGS to model VO/OV order in Latin and Old French
 - Levshina (2016) used RStan to model the use of three near-synonymous constructions (let/allow/permit + (to) Infinitive)

Communicative efficiency

- When reporting the results of traditional maximum likelihood regression, linguists seldom add "frequentist", "ML" or some other attribute.
- When reporting results of Bayesian regression, they normally use "Bayesian".
- Frequentist regression is still the default (unmarked), whereas Bayesian regression is a deviation (marked).
- Cf. in German *Auto* vs. *E-Auto*...

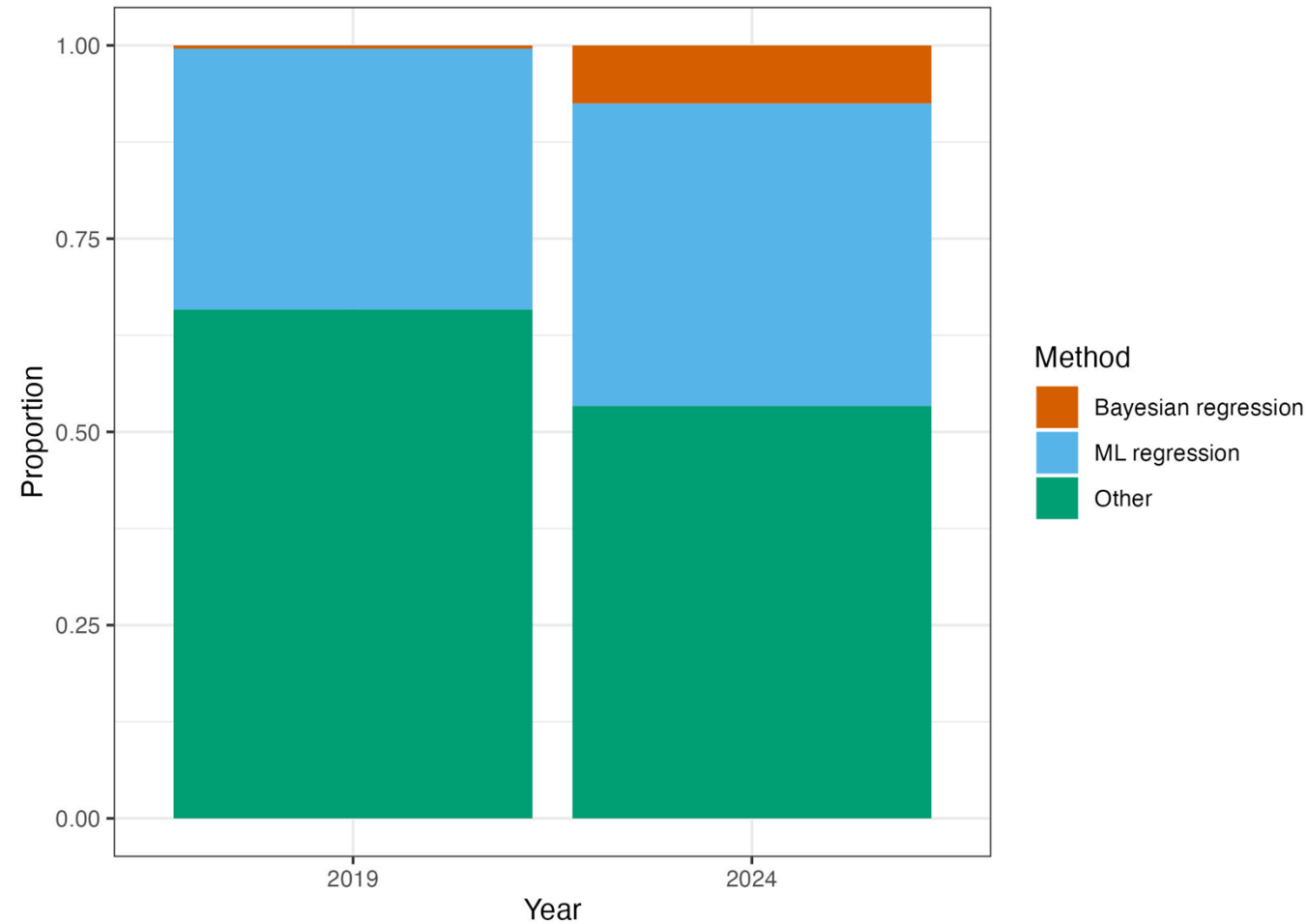
Quantitative turn in linguistics

- Quantitative approaches were always present (*Zipf 1935; see also Joseph 2008*)
- Kortmann (2021): the quantitative turn in linguistics between the end of the 20th and the beginning of the 21st century.
- Janda (2013): 2008 was a pivotal year in the history of Cognitive Linguistics: the majority of articles contained quantitative analysis of authentic language data after 2008.
- But has there been a Bayesian turn?



A survey

Bayesian, Maximum Likelihood regression and other methods in ten linguistics journals in 2019 and 2024.



Linguistics journals in 2024

Journal	N papers checked	N papers reporting regression	N papers reporting Bayesian regression
Cognitive Linguistics	19	11	4
English Language and Linguistics	24	6	1
Intl. Journal of Corpus Linguistics	20	3	0
Journal of Phonetics	22	20	3
Journal of Pragmatics	25	4	0
Language	22	11	2
Language and Cognition	50	14	2
Language Variation and Change	15	13	3
Linguistic Typology	13	3	2
Linguistics	43	15	2
Total	253 (100%)	118 (46.6%)	19 (7.5%)

Or 16% of all
regression papers 39

Causes of shift

- Increasingly complex models and research questions
- Software (brms, rstan, rjags, rstanarm, JASP, bma, bms...) with excellent documentation
- Friendly Stan community
- Books and tutorials – general, for cognitive scientists & linguists (Kruschke 2015, Nicenboim & Vasisht 2016, Sorensen et al. 2016, Kimball et al. 2019, Levshina 2022...)

Tricky cases

- Maximal random effects structure
- Multiple categories in the response variable
- Zero/one inflated data
- Phylogenetic regression
- Too many predictors (large P) and multicollinearity
- Small data (small N)
- Missing data

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Maximal random effects structure

- Barr et al. (2013): Keep it maximal!
- Husain et al. (2014): maximal models did not converge in lme4, but were possible to fit in Stan:

$$y_{jk} = (\beta_0 + b_{u0j} + b_{w0k}) + (\beta_1 + b_{u1j} + b_{w1k})A_{jk} \\ + (\beta_2 + b_{u2j} + b_{w2k})B_{jk} + (\beta_3 + b_{u3j} + b_{w3k})A_{jk} \times B_{jk} + \epsilon_{jk}$$

- Maximal Bayesian models are still popular in psycholinguistics (e.g. *Copot & Bonami 2024 CogLing; Scholman et al. 2024 Glossa Psycho; Winkowski et al. 2024 Glossa Psycho*).
- One can use regularizing priors on the correlations between random effects to make the estimates more conservative.

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Categorical and ordinal responses

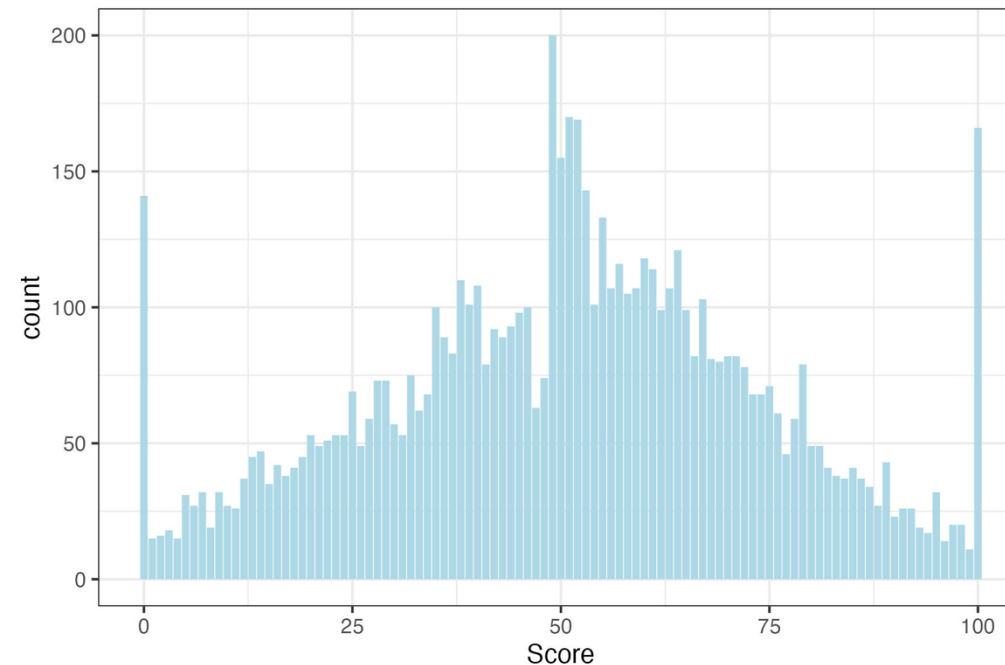
- Levshina (2016) used Stan because of the practical difficulties of using traditional mixed-effects logistic regression with more than two categories in the response variable back then.
 - Now: the 'categorical' family in brms.
- Several options for ordinal variables:
 - 'cumulative' family is the best for Likert-item datasets, in which ordered categories (presented as numbers or words) may reflect an underlying continuous psychological variable
 - Examples: Winkowski et al. (2024), Zhang & Davidson (2024) in *Glossa Psycho*

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Slider data

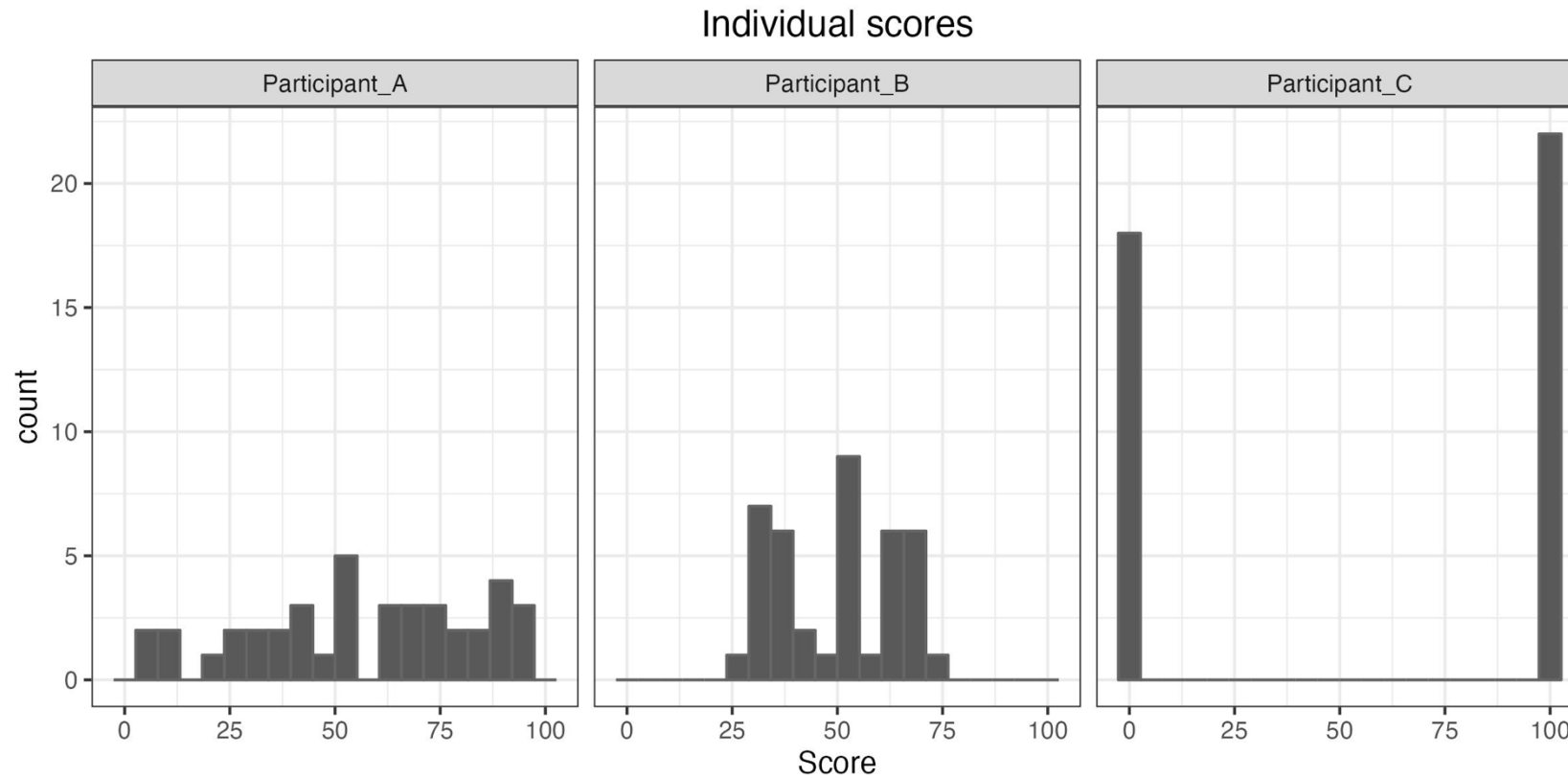
- de Hoop et al. (2023): a 100-point slider scale to measure addressees' responses to emails written in the informal pronoun T or the formal pronoun V in Dutch HR communication.



Zero-one inflated beta regression

- a traditional beta regression of the scores between 0 and 1;
- a model of the so-called centrality parameter ϕ , which shows how variable the data are;
- a logistic zero-one inflated model predicting 0 or 1 vs. all other scores;
- a logistic model predicting 1 vs. 0, given that the outcome is either 1 or 0.

Modelling individual variation



de Hoop et al. 2023 J of Pragmatics

$\phi_i \sim (1 | \text{ParticipantID}_i)$, $z_{oi} \sim (1 | \text{ParticipantID}_i)$

Results

Model	Mean posterior of Pronoun	Lower boundary of 95% Credible Interval	Upper boundary of 95% Credible Interval	Posterior probability of positive effect
Gaussian	0.90 (on original scale)	0.06	1.74	100%
beta with modified scores	0.04 (on logit scale)	0.01	0.07	99.3%
zero-one-inflated (ZOIB) beta	0.03 (on logit scale)	0.00	0.06	98.6%

de Hoop et al. 2023 J of Pragmatics

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Who did what to whom?



(man, dog, bite)

Images from <https://dogdailynews.wordpress.com/2009/09/28/man-bites-dog/>,
<https://www.seagull-tandem.eu/portfolio/b1-dog-bites-man/>

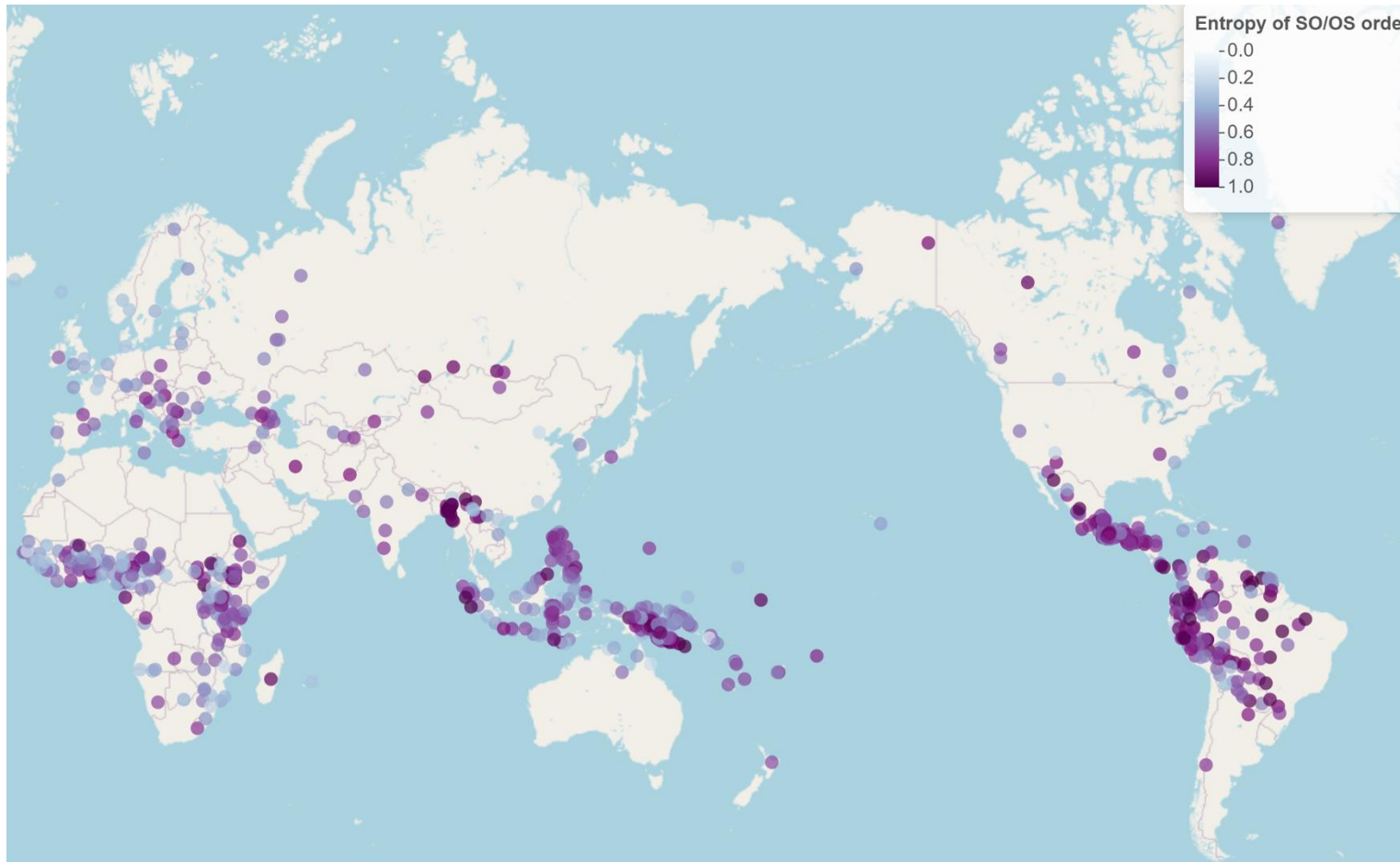
Cues to Subject and Object

- Case and agreement (German, Latin, Russian, Spanish)
- Rigid order of core arguments (English, Mandarin Chinese)
- Semantics: categorical restrictions or probabilistic constraints on semantic classes or more specific nouns (*e.g., Aissen 2003, Kurumada & Jaeger 2015*)
- POS, person, information status (*Levshina 2021 Ling Van*)
- Verb-medial position helps to recognize Subject and Object in noisy communication (*Gibson et al. 2013*).

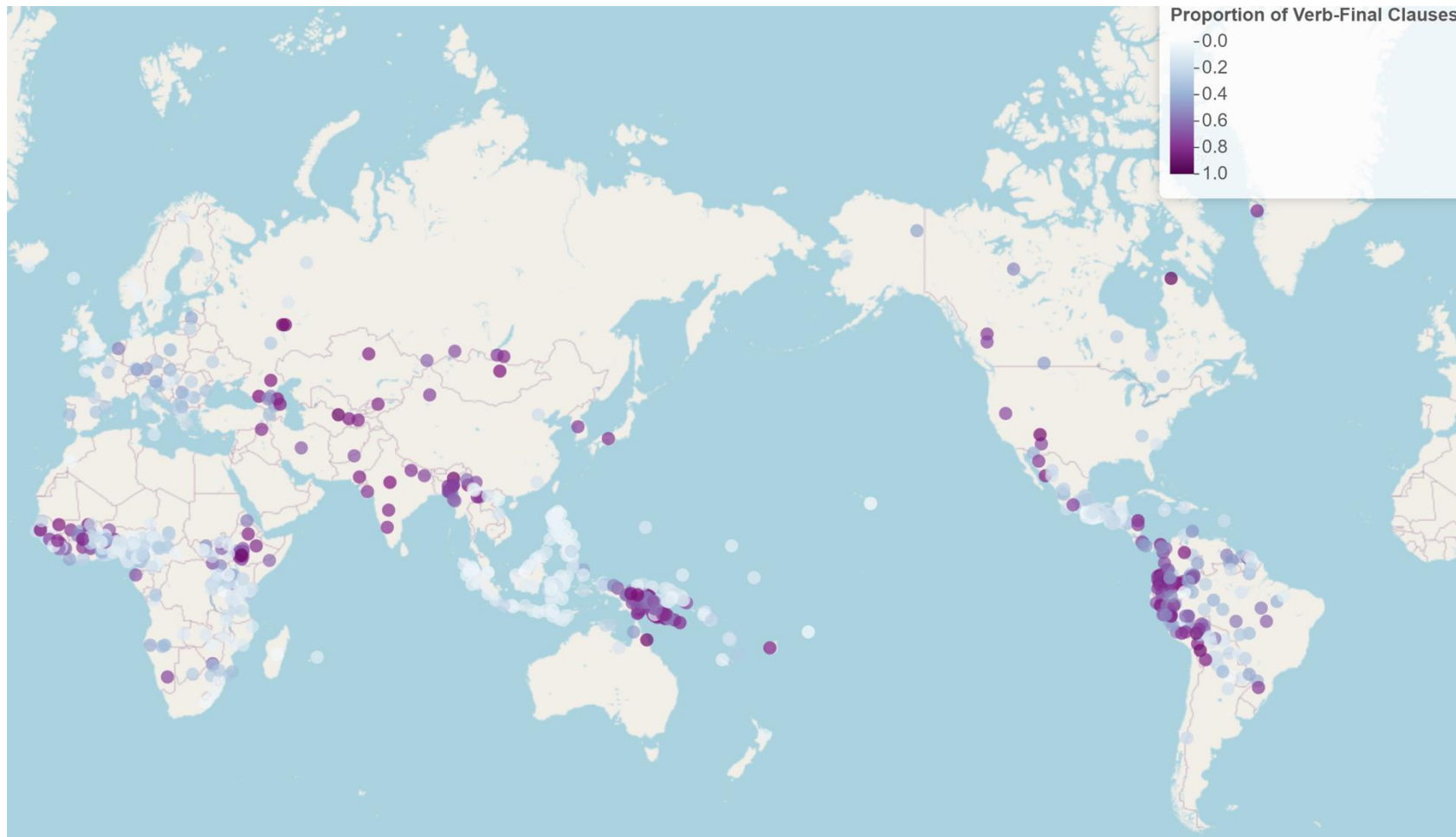
Data

- Parallel Bible Corpus (Mayer & Cysouw 2014), New Testament translations
829 languages from 99 families
- Multilingual alignment word order data from Östling (2015)
- Number of speakers from Ethnologue

SO/OS order entropy in New Testament



Proportions of Final Clauses in New Testament



Levshina, In prep.

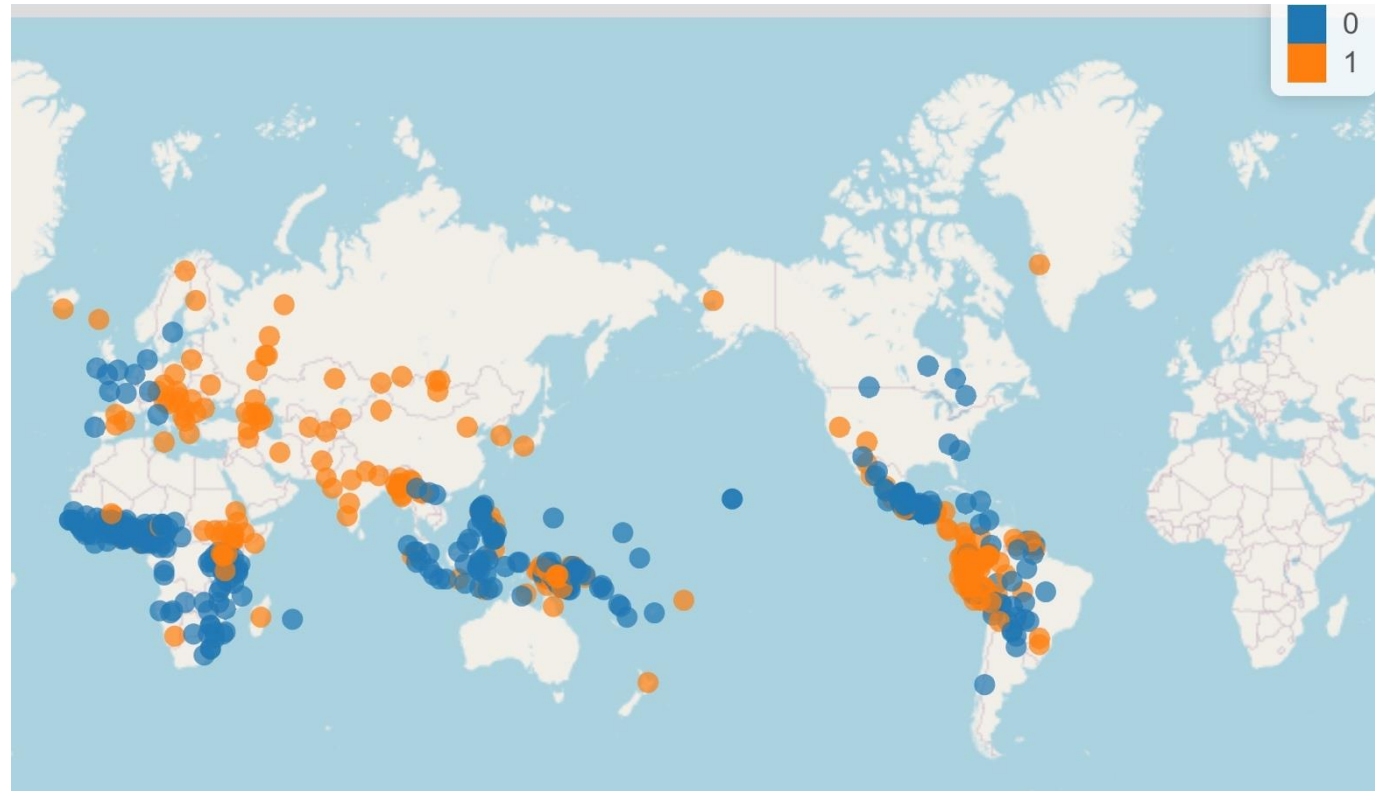
Case forms of S and O

0 = identical forms

1 = different forms

- morphological case
- adpositions
- particles
- clitics
- tones

Restricted/differential marking counts as 1!



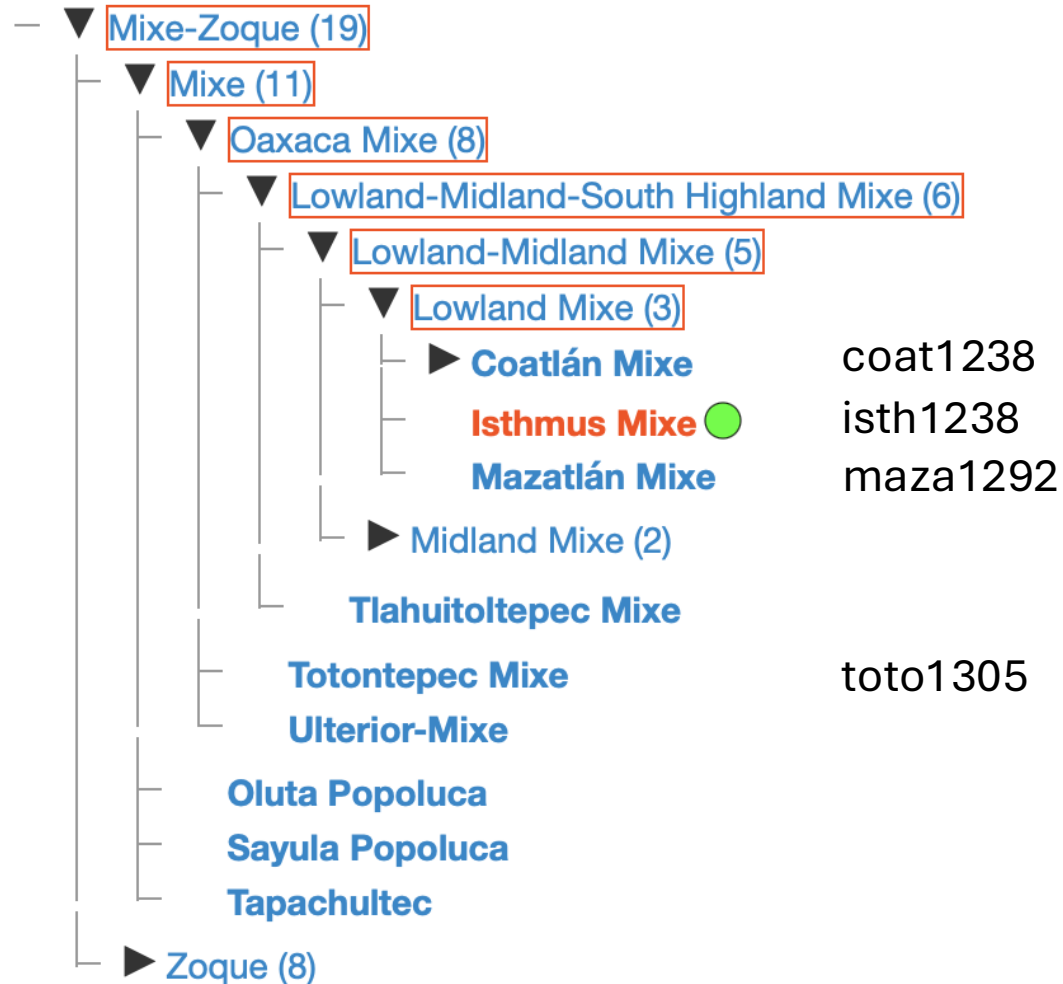
WALS + Grambank (partly) > reference grammars >
extrapolation for Algonquian, Atlantic-Congo, Mande, Mayan and
Turkic languages

Levshina, In prep.

Hierarchical phylogenetic regression

- Originally developed in evolutionary biology (*Hadfield & Nakagawa 2010*)
- Convenient implementation in brms (*Bürkner 2024*)
- Random structure reflects correlations between species (langoids belonging to different genera and families).

Example: Mixe-Zoque languages



Phylogenetic regression models

P (beta > 0)

Response ~ Predictor	Rigid order	Verb final	Case	N speakers
Rigid order		0.00	0.00	0.77
Verb final	0.00		1.00	0.07
Case	0.01	1.00		0.84
N speakers	0.60	0.15	0.85	

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Too many predictors

- How to choose which ones are useful and avoid overly complex models?
- Frequentist models: LASSO, ridge, elastic net...
- Bayesian approach: shrinkage priors
 - E.g. horseshoe priors with a global shrinkage hyperparameter that shrink all predictors to zero and a local shrinkage hyperparameter that allows some coefficients to escape the shrinkage (*Carvalho et al. 2009*).
 - Convenient to use in brms.
 - Also helps with multicollinearity
- Another Bayesian approach: Bayesian model averaging or stacking (*Kaplan 2021*)
 - computing posterior model probabilities and identifying the most likely models
 - posterior inclusion probabilities for every predictor
 - Packages brms, BMA and BMS in R.

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Small data and strong informative priors

- Levshina (2022): a comparison of frequentist and Bayesian logistic models of help (to) Infinitive
- A large sample (2300 examples): the priors do not really play a role.
- A smaller sample (400 examples) annotated for a time-consuming variable:
 - Informative priors based on the large sample
 - Testing the effect of the new variable and avoiding technical issues due to data sparseness.



https://www.academia.edu/57805622/Comparing_Bayesian_and_frequentist_models_of_language_variation_The_case_of_help_to_Infinitive_final_version_

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Missing data

- brms can be used to model missing or imputed data and data with measurement error
- You can first impute the values (e.g., by using the mice package), fit a brm_multiple model and then combine the posterior samples.
- You can also use imputation directly during model fitting.
- See vignette "brms_missings".
- See also <https://bookdown.org/content/4857/missing-data-and-other-opportunities.html#measurement-error>

A bonus: Testing the Null Hypothesis

- Dienes 2014: "Using Bayes to get the most out of non-significant results"
- Your p -value is > 0.05 .
- Is it due to true lack of effect (true null hypothesis) or something else?
- Compute Bayes Factor!
- Fewer studies will end up in a drawer.

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Why bother?

- In most cases, the results of frequentist and Bayesian regression will be similar.



Conceptual advantages of Bayesian statistics

- More intuitive (cf. credible intervals)
- Everything is probabilistic → greater awareness of the arbitrary nature of any cutoffs
- Integration of prior information (based on previous research and common sense)
- Ways of testing the null hypothesis against the alternative → mitigating the publication bias, no "null results" ending in the drawer.

The dark side of Bayesian statistics

- Computationally demanding
- Some tools (e.g., Bayes Factors) are prone to misinterpretation and misuse, e.g., BF-hacking:
 - ". . . surely, God loves the .06 nearly as much as the .05" (*Rosnow & Rosenthal 1989: 1277*)
 - "God would love a Bayes Factor of 3.01 nearly as much as a BF of 2.99." (*Konijn et al. 2015*)
- More diagnostics (MCMC chain convergence): R-hats, effective sample sizes, autocorrelations...
- Too many "degrees of freedom": Which priors? HDI or ETI? 95% or 89%? ROPE, PD, BF?

Ask yourself!

- Do you often have to deal with tricky cases like the ones mentioned?
- Do researchers in your field use Bayesian regression?
- Are journal editors in your field sympathetic towards Bayesian methods?
- Do you enjoy learning cutting-edge methods?
- Are you an active reviewer of papers that employ different types of regression – now or in the future?
- Do you want to contribute to cumulative science?



Bayesian or frequentist?

- Denes (2011) puts a question in his provocative paper, “Bayesian Versus Orthodox Statistics: Which Side Are You On?”
- I personally don’t think this is a good question. Instead:

“A spirit of plurality is needed that creates space for realizing that there are problems and applications that might be handled more easily by either Bayesian or frequentist approaches, and that the analyst is far better off having both tools in their toolbox. In particular, researchers need to be familiar with both approaches, since there is an increasing number of papers in quantitative linguistics that uses Bayesian approaches.”